1(a)	$\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right) \ldots x^{1}+\ldots x^{-\frac{3}{2}}+\ldots x^{-2}$	M1
	$\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right) \ \frac{2}{3}x \ -2x^{-\frac{3}{2}} \ -\frac{8}{3}x^{-2}$	A1A1A1
		(4)
(b)	$\frac{\mathrm{d}y}{\mathrm{d}x}="\frac{2}{3}"\times 4-"2"\times 4^{"-\frac{3}{2}"}-"\frac{8}{3}"\times 4^{"-2"}=\frac{9}{4}$	M1
	$\frac{9}{4}\rightarrow -\frac{4}{9}$	M1
	$y-3="-\frac{4}{9}"(x-4)$	dM1
	$4x+9y-43=0$	A1
		(4)
		(8 marks)

	$-a+6a+8+a^2=32 \Rightarrow a^2+5a-24=0$	M1
	$(a+8)(a-3)=0$	dM1
	$a=3$ or $a=-8$ and chooses $a=3$ with reason *	A1* cso
		(3)
(ii)	$3x^3+26x^2-9x=0 \Rightarrow x(3x^2+26x-9)=0$	M1
	$x(3x-1)(x+9)$	
	$(x=)\ 0, \frac{1}{3}, -9$	A1
		(2)
(b)(i)	$(y=)\ 0$	B1
	$y^{\frac{1}{3}}="\frac{1}{3}"$ or $y^{\frac{1}{3}}="-9" \Rightarrow y=...$ (or $(-9)^3=...$ or $\left(\frac{1}{3}\right)^3=...$)	M1
	$(y=)\ \frac{1}{27},\ -729$	A1
		(3)
(b)(ii)	$9^z="\frac{1}{3}" \rightarrow z=...$	M1
	$(z=)\ -\frac{1}{2}$ only	A1
		(2)
		(10 marks)

3(a)	$\left(2\sqrt{2}\right)^2 = p^2 + q^2 - 2pq\cos 60°$ oe $p^2 + q^2 - pq = 8$ *	M1 A1*
		(2)
(b)	$q = p + 2 \Rightarrow 8 = p^2 + ("p+2")^2 - p("p+2")$	M1
	$p^2 + 2p - 4 = 0$ or $q^2 - 2q - 4 = 0$	A1
	$p = \frac{-2 \pm \sqrt{2^2 - 4\times 1\times(-4)}}{2}$ or $q = \frac{2 \pm \sqrt{(-2)^2 - 4\times 1\times(-4)}}{2}$	M1
	$p = -1 + \sqrt{5}$ or $q = 1 + \sqrt{5}$	B1 (A1 on EPEN)
	$p = -1 + \sqrt{5}$ and $q = 1 + \sqrt{5}$ only	A1cso
		(5)
(c)	$\text{Area} = \frac{1}{2} \times ("-1+\sqrt{5}")("1+\sqrt{5}") \times \sin 60°$	M1
	Area = $\sqrt{3}$ (m^2)	A1
		(2)
Alt(a)	Forming a line BX which is perpendicular to AC where X is on the line AC.	
	$AX = p\cos 60 = \frac{p}{2}$	
	$BX = \sqrt{p^2 - \left(\frac{p}{2}\right)^2} = \frac{\sqrt{3}}{2}p$ or $BX = p\sin 60$	
	$\left(\frac{\sqrt{3}}{2}p\right)^2 + \left(q - \frac{p}{2}\right)^2 = \left(2\sqrt{2}\right)^2$ or $(p\sin 60)^2 + \left(q - \frac{p}{2}\right)^2 = \left(2\sqrt{2}\right)^2$	M1
	$\frac{3p^2}{4} + q^2 - pq + \frac{p^2}{4} = 8$	
	$p^2 + q^2 - pq = 8$ *	A1*
		(9 marks)

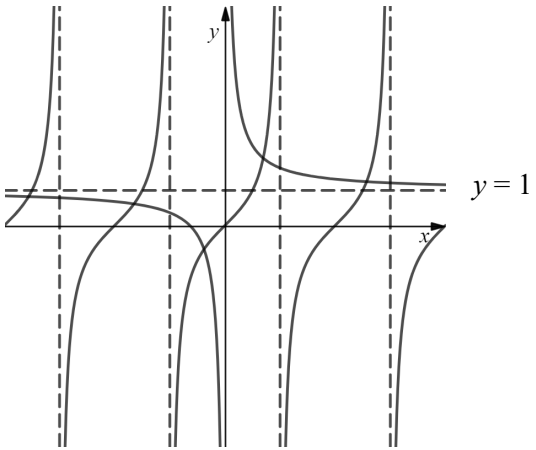
4	$\int \frac{3x^{\frac{3}{2}} - 15x^{\frac{1}{2}} + 2x - 10}{4\sqrt{x}} \mathrm{d}x = \int \frac{3}{4}x - \frac{15}{4} + \frac{1}{2}x^{\frac{1}{2}} - \frac{5}{2}x^{-\frac{1}{2}} \mathrm{d}x$	M1A1A1
	$x^n \rightarrow x^{n+1}$	dM1
	$\frac{3}{8}x^2 - \frac{15}{4}x + \frac{1}{3}x^{\frac{3}{2}} - 5x^{\frac{1}{2}} + C$	A1A1
		(6 marks)

5(a)	$P_B - P_A = 44.2 - \left(53 - 0.4 \times 8^2\right) = ...$ awrt (£) 16.8 million	M1 A1
		(2)
(b)	(£) 53 (million)	B1
		(1)
(c)	$-1.6t + 44.2 = 53 - 0.4(t-8)^2$ $\Rightarrow 0.4t^2 - 8t + \frac{84}{5} = 0 \Rightarrow t = ...$ $t = 10 - \sqrt{58} =$ awrt 2.38 (years) "2.38" $< t \left(\leqslant 15\right)$	M1 M1 A1 A1ft
		(4)
(d)	"The **share value** would be negative" / "the **model** is known to hold for 15 years only (and 20 years is more than 15)."	B1
		(1)
		(8 marks)

6(a)	$\mathrm{f}'(8)=\dfrac{32}{3\times 8^2}+3-2\sqrt[3]{8}\quad\left(=-\dfrac{5}{6}\right)$	M1
	$y-2="-\dfrac{5}{6}"(x-8)$	dM1
	$y=-\dfrac{5}{6}x+\dfrac{26}{3}$	A1
		(3)
(b)	$\mathrm{f}'(x)=\dfrac{32}{3x^2}+3-2\sqrt[3]{x}=...x^{-2}+3+...x^{\frac{1}{3}}$	
	$x^{-2}\to x^{-1},\ 3\to 3x,\ x^{\frac{1}{3}}\to x^{\frac{4}{3}}$	M1
	$\mathrm{f}(x)=\int\dfrac{32}{3}x^{-2}+3-2x^{\frac{1}{3}}\mathrm{d}x=-\dfrac{32}{3}x^{-1}+3x-\dfrac{3}{2}x^{\frac{4}{3}}+c$	A1A1
	$2=-\dfrac{32}{3}\times 8^{-1}+3\times 8-\dfrac{3}{2}\times 8^{\frac{4}{3}}+c\Rightarrow c=...$	dM1
	$\left(\mathrm{f}(x)=\right)\ -\dfrac{32}{3}x^{-1}+3x-\dfrac{3}{2}x^{\frac{4}{3}}+\dfrac{10}{3}$	A1
		(5)
		(8 marks)

7(a)	y coordinate $= 12$	B1
		(1)
(b)	Gradient of $l_1 = -\frac{3}{4}$	B1
	$\Rightarrow$ Gradient of $l_2 = \frac{4}{3} \Rightarrow (y-6) = "\frac{4}{3}"(x-8)$	M1
	y coordinate $= -\frac{14}{3}$ *	A1* cso
		(3)
(c)	Radius $= "12" + \frac{14}{3} = \frac{50}{3}$	B1ft
	Length of arc $= "\frac{50}{3}" \times 1.8 = 30$	M1A1cao
		(3)
(d)	Area of sector $= \frac{1}{2} \times ("\frac{50}{3}")^2 \times 1.8 \quad (= 250)$	M1
	$"250" + \frac{1}{2} \times "\frac{50}{3}" \times 8 = "250" + "\frac{200}{3}"$	M1
	$= \frac{950}{3}$ (units^2)	A1cao
		(3)
		(10 marks)

8(a)	$3x^2+6x+9=3(x\pm...)^2\pm...\qquad a=3$	B1
	$3x^2+6x+9=3(x+1)^2\pm...\qquad a=3\ \&\ b=1$	M1
	$3x^2+6x+9=3(x+1)^2+6$	A1
		(3)
(b)	$(-1,\ 6)$	B1ft
		(1)
(c)	$y=\alpha(x+4)(x+2)(x-3)$	B1
	$6=\alpha(-1+4)(-1+2)(-1-3)$	M1
	$\alpha=-\frac{1}{2}$	A1
	$y="-\frac{1}{2}"(x+4)(x+2)(x-3)\Rightarrow y=...x^3+...x^2+...x+...$	M1
	$A=-\frac{1}{2},\ B=-\frac{3}{2},\ C=5,\ D=12$	A1
		(5)
Alt (c)	$-64A+16B-4C+D=0$ $-8A+4B-2C+D=0$ $27A+9B+3C+D=0$	B1
	$-A+B-C+D=6$	M1
	One of $A=-\frac{1}{2},\ B=-\frac{3}{2},\ C=5,\ D=12$	A1
	Fully solves their simultaneous equations	M1
	$A=-\frac{1}{2},\ B=-\frac{3}{2},\ C=5,\ D=12$	A1
		(9 marks)

9(a)	$x = \frac{3\pi}{2}$ oe	B1
		(1)
(b)(i)		B1B1
(ii)	5 (solutions)	B1
	Number of solutions are the number of points of intersections between the graphs	B1
		(4)
(c) **(i)**	(Number of solutions) = 40	B1ft
(ii)	(Number of solutions) = 14	B1
		(2)
		(7 marks)